

HAT-P-19b

R-1016

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_241219 · created 2026-07-30T03:48:46+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460663.7451 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.014
Transit depth (Rp/Rs)^2	2.65 +/- 0.46 [%]
Orbital Inclination (inc)	87.64 +/- 0.63
Airmass coefficient 1 (a1)	0.986 +/- 0.015
Airmass coefficient 2 (a2)	0.0067 +/- 0.004
Scatter in the residuals of the lightcurve fit is	1.55 %
Best Comparison Star	#3 - [429, 316]
Optimal Aperture	3.34
Optimal Annulus	13.24
Transit Duration (day)	0.106 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.163 ± 0.014 vs 0.1418 (deviation 1.51σ)
Duration fit vs published	0.106 d vs 0.1196 d (deviation 1.36σ)
Mid-time O–C	1.8 min
Depth SNR	11.6
Residual scatter	1.55 %

Light curve

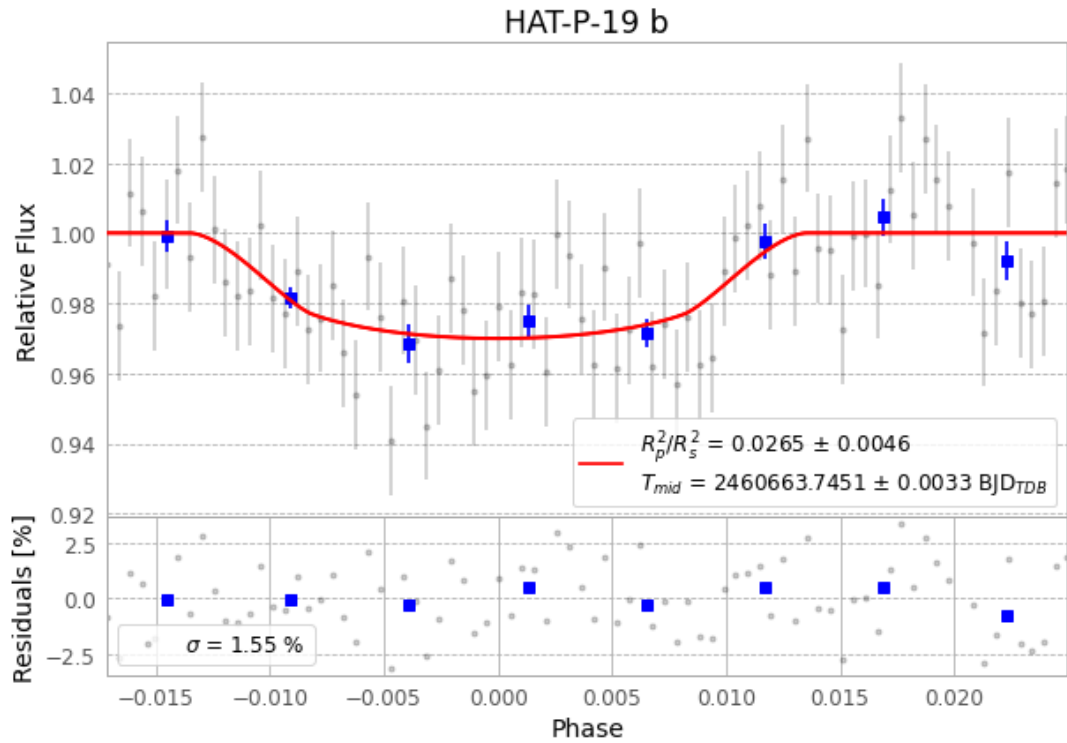


Figure 1 — FinalLightCurve_HAT-P-19 b_2024-12-18.png (SHA-256 59567e1051fca7eb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 342c021d78cff6a4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

82 FITS frames (54 MB), HATP-19241219040916.FITS → HATP-19241219081215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-241219.log (b614966a28eb...), FinalLightCurve_HAT-P-19 b_2024-12-18.png (59567e1051fc...), FinalLightCurve_HAT-P-19 b_2024-12-18.csv (076f689abe7b...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 03:48Z.